Numerical study of Magnetohydrodynamics for turbulent flow using variational multiscale method

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Dedicated to my Parents

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Abstract

The goal of the project is to capture the dynamics of a Magnetohydrodynamic (MHD) flow of high Reynolds number and its effect on heat transfer using projection based variational multiscale (VMS) finite element method. The coupled Maxwell and Navier-Stokes equations with quasi-two-dimensional (Q2D) approximation is solved to study the behavior of a 3D MHD flow in a duct with a square cross-section and a square obstacle. The efficiency of VMS in MHD with Q2D approximation problem is studied. To effect on heat transfer due to the external magnetic field is studied by calculating time averaged Nusselt number. All the studies are done for Hartmann number from 320 to 2160 and Reynolds number from 100 to 10000.

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Chapter 1

Introduction

1.1 Motivation

From the theory of electrodynamics, it is well known that the motion of an electrically conducting material across magnetic field lines induces electric currents in the conductor. This principle of electromagnetic induction as been used in the variety of engineering applications such as an electrical generator, electrical transformer, magnetic braking in trains. Apart from this, the theory has been used widely in Magnetohydrodynamics (MHD) problem related to geophysics and astrophysics to study the behavior of plasma in Earth's core and in stars. MHD is currently being used at industry and laboratory scales to study the behavior of liquid metal flow under the presence of external magnetic fields. Most of the studies are focused on modeling the behavior of self-cooled liquid metal fusion blanket in fusion reactors such as International Thermonuclear Experimental Reactor (ITER) Fig.(1.1) and liquid metal fast breeder reactors (LMFBR). As explained by Phillip et al. [5], in ITER, under magnetic confinement fusion approach, an external magnetic field is used to confine the hot fusion fuel in the form of a plasma. In these reactors, the energy production comes from the high-energy neutrinos, electrons, and photons that are ejected from the fusion reaction that hit a blanket surrounding the containment vessel, thus heating the blanket. The blanket is cooled by a liquid metal coolant which in turn transfers the heat to a steam turbine energy generator. The interaction of this coolant liquid metal with intense magnetic fields is being analyzed to keep the temperature at

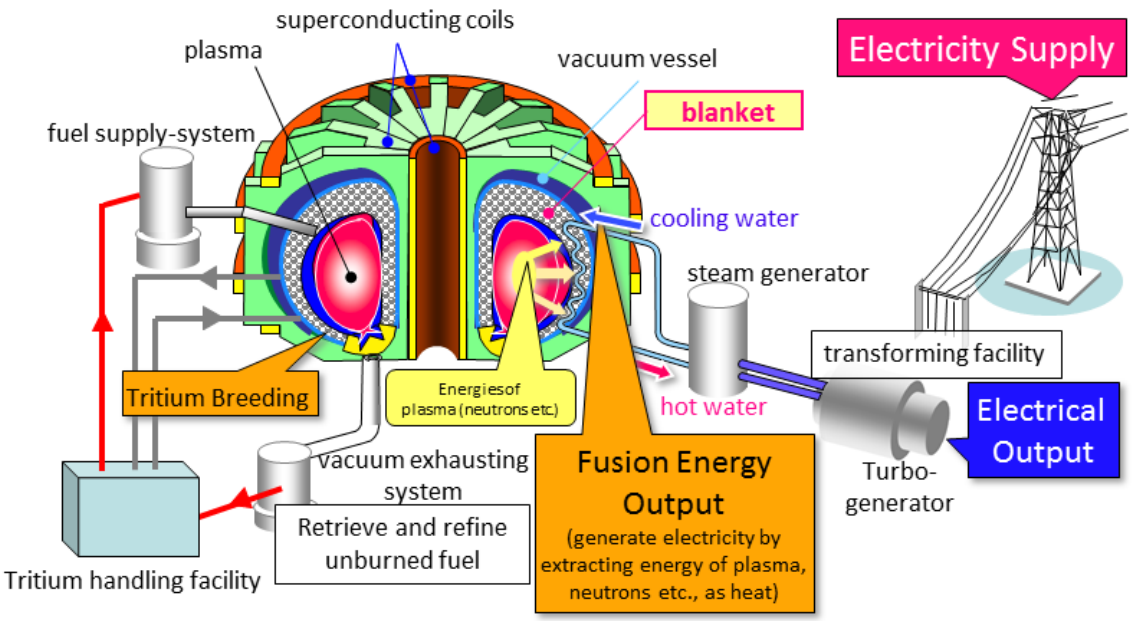

Figure 1.1: Schematic Diagram of The International Thermonuclear Experimental Fusion Reactor (ITER), taken from [1].

the plasma facing wall within acceptable limits.

Usually turbulence of flow plays a major role in heat transfer, hence less turbulence can affect the thermo-fluidic transport significantly. In the fusion reactor, liquid metal blankets tend to laminarize due to magnetic damping. When a electrical conducting liquid metal passes through a duct, as shown in Fig.(1.2) , in presence of imposed magnetic field, eddy currents are created which in turns creates a drag force, usually called as Lorentz force, in direction opposite to the flow. This damping effect reduces the flow turbulence, which leads to degradation of heat transfer. The heat transfer can be augmented by placing an obstacles inside the blankets, as shown in Fig.(2.1) to increase the turbulence. The obstacles may have different geometric shapes such as circular or square cross-sections and the thermo-fluidic transport around the obstacles will strongly depend on the shape of the obstacle.

Various papers have discussed the heat transfer of MHD flow but have restricted the study to 6000 to 7000 Reynolds number. Various papers have discussed the heat transfer in MHD flows

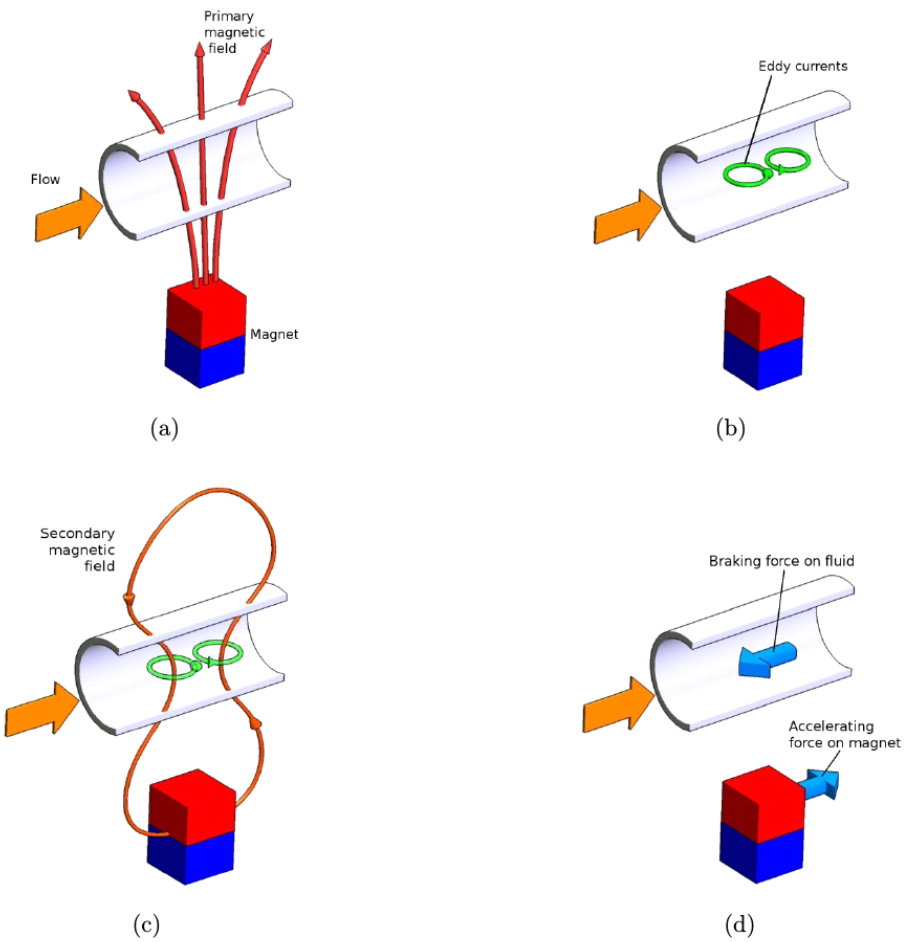

Figure 1.2: Schematic of the working principle of Lorentz force, taken from [2].

with a range of Reynolds numbers from 6000 to 7000 using finite volume, finite difference method or finite element method. Finite element methods have gained immense popularity owing to its efficiency in solving partial differential equations for complex geometries. In the current study, the turbulent magnetohydrodynamic flow and its effect on heat transfer are investigated using finite element (FE) based approach. To this end, a quasi-2D based MHD model proposed by Sommeria [6] famously known as the SM82 model is utilized. The SM82 model is simple, easy to implement and can lead to a dramatic gain in CPU time in comparison with a fully 3D numerical simulation. Further, the SM82 model can be efficiently used to acquire extensive information for a wide variety aspects of the flow.

There are many known numerical techniques to solve the problem but we have used finite element method (FEM). It is preferred owing to its efficiency in solving the differential equation for complex geometries. It reduces the given problem into systems of algebraic equations, generally linear in nature, that need to be solved. However, turbulent flows are highly unsteady, where the main velocity field is superimposed by random fluctuations of different scales which can not be simulated by direct discretizations of equations Navier-Stokes equations, for e.g. by a Galerkin finite element. For this purpose variational multiscale method has been implemented as proposed in the literature, see for example [7–14]

1.2 Outline

The structure of the thesis is as follows. Chapter 2 gives the Introduction to Magnetohydrodynamics and assumptions behind the SM82 model. In Chapter 3 the Introduction to the Variational multiscale method is given. Chapter 4 discusses the mathematical formulations using finite element technique. In Chapter 5 Numerical results are discussed. The conclusions are presented in Chapter 6.

Chapter 2

Magnetohydrodynamics

2.1 General Magnetohydrodynamics equation

Magnetohydrodynamics (MHD) is a single fluid macroscopic non-relativistic theory that is concerned with the phenomenon in magnetic plasma or metallic fluids. It was first initiated by Hannes Alfvén for which he received Nobel prize in 1970. It has a wide range of applications, from natural phenomenon like geophysics, astrophysics to engineering applications such as electromagnetic casting, liquid metal cooling in nuclear reactors and so on. The fundamental equations to describe quasi-neutral Magnetohydrodynamics phenomenon are:

Equations of electromagnetism

$$\nabla \cdot \mathbf{D} = \lambda \qquad \text{Coulombs law} \tag{2.1a}$$

$$\nabla \cdot \mathbf{B} = 0 \qquad \text{Gauss law} \tag{2.1b}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \text{Faraday's law} \tag{2.1c}$$

$$\nabla \times \mathbf{B} = \mu \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \qquad \text{Ampere's law} \tag{2.1d}$$

$$\mathbf{J} - \mathbf{R} = \sigma \left(\mathbf{E} + \mathbf{u} \times \mathbf{B} \right) + \lambda \mathbf{u} \qquad \text{Generalized Ohm's law} \tag{2.1e}$$

$$\nabla \cdot \mathbf{J} = -\frac{\partial \lambda}{\partial t} \qquad \text{Current continuity} \tag{2.1f}$$

and Navier Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} - 2\nu\nabla\cdot\mathbb{D}(\mathbf{u}) + \mathbf{u}\cdot\nabla\mathbf{u} + \frac{1}{\rho}\nabla p = \frac{1}{\rho}\mathbf{f_b} + \frac{1}{\rho}\mathbf{f_e} \qquad \text{Momentum Conservation} \tag{2.1g}$$

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\mathbf{u}) = 0 \qquad \text{Continuity equation} \tag{2.1h}$$

Where, μ is magnetic permeability, σ is electrical conductivity, **D** is electric flux density, λ is charge density, **B** is magnetic flux density, **E** is electric field, **J** is electric current density, **u** is fluid velocity, p is the pressure, ρ is fluid density, ν is Kinematic Viscosity, $\mathbf{f_b}$ is body force. The $\mathbf{f_e}$ and the velocity deformation tensor, $\mathbb{D}(\mathbf{u})$ are expressed as given by :

$$\mathbb{D}(\mathbf{u}) = \frac{\nabla\mathbf{u} + \nabla\mathbf{u}^T}{2}$$

$$\mathbf{f_e} = \lambda\mathbf{E} + \mathbf{J}\times\mathbf{B}$$

The term **R** in equation (2.1e) contains Hall effect, electron pressure force and electron inertia. The problems related to MHD are usually studied by simultaneously solving Navier-Stokes equations and Maxwell's equations of electromagnetism. To define any such MHD model nondimensional parameters are $Re,\ N,\ Ha$ and Rm are used which are defined as:

- Reynolds number is ratio of inertial forces to viscous forces within a fluid and given by,
$$Re = \frac{UL}{\nu}$$
- Stuart number N, is defined as ratio of lorentz force and inertia force. It is calculated as
$$N = \frac{\sigma a B^2}{\rho U}$$
- Hartmann Number Ha, is the ratio of Lorentz force and viscous force. It is calculated as
$$Ha = aB\sqrt{\frac{\sigma}{\rho\nu}}$$
- Magnetic Reynolds number Rm, the ratio of the time scales of diffusion $\tau_d = L^2\mu\sigma$ and induction $\tau_a = L/U$ of magnetic field given by
$$Rm = \sigma\mu UL$$

L, U denotes the characteristic length and velocity respectively. a is the dimension of duct in direction of magnetic field as shown in Fig.(2.1). In order to simulate models at industry and laboratory scales above equations are simplified with little error using following assumptions.

- Fluid velocity is much smaller than the speed of light. Hence, the non-relativistic equations are valid. The assumption is usually referred as the low-frequency problem.
- Velocity of the charge carriers (electrons or ions) remains small with respect to fluid velocity and the charge carriers move within the fluid without inertia
- In many cases, the induction-less magnetic field is assumed for very low magnetic Reynolds number, Rm

As a result of these assumptions, the charge relaxation time defined as $\tau_r = \epsilon/\sigma$ is very small, usually of order of 10^{-18}. The Equations (2.1) is simplified as follows:

- In generalised ohm's law, **R** is very small as compared to other terms. In the frame of the quasi-static approximation, the convection current $\lambda\mathbf{u}$ can then be neglected with respect to $\sigma\mathbf{E}$ as the ratio $\lambda\mathbf{u}/\sigma E \sim \tau_r U/L$. The final Ohm's law equation becomes
$$\mathbf{J} = \sigma\left(\mathbf{E} + \mathbf{u} \times \mathbf{B}\right)$$
- Similarly in Eq(2.1f), on substituting ohm's law the equation becomes:
$$\partial_t \lambda + \sigma \nabla \cdot \mathbf{E} + \nabla \cdot (\mathbf{u} \times \mathbf{B}) = 0$$
In this equation, $\sigma \nabla \cdot \mathbf{E} \sim q/\tau_r$. Therefore last term, $\nabla \cdot (\mathbf{u} \times \mathbf{B})$ can be neglected which leads us to
$$\partial_t \lambda + \frac{q}{\tau_r} = 0$$
$$\therefore \quad \nabla \cdot \mathbf{J} \approx 0$$
- In ideal case i.e. for perfect conductor, $\sigma \to \infty$
$$\mathbf{E} = -\mathbf{u} \times \mathbf{B}$$
$\mathbf{E} \sim UB$. When we consider in ampere's law of equation (2.1d) we compare
$$\nabla \times \mathbf{B} \sim \frac{B}{L} \quad \text{and} \quad \mu \frac{\partial \mathbf{D}}{\partial t} = \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} \sim \frac{E}{c^2 t_o} \sim \frac{U^2}{c^2}\frac{B}{L}$$
For $U << c$ the induction current is negligible. Therefore we obtain
$$\nabla \times \mathbf{B} = \mu \mathbf{J}$$
- Rm at industrial scale is very small. Consider equation (2.1c) and modified equations

(2.1e), (2.1d) after the above assumptions. On combining the equations we obtain following induction equation

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \frac{1}{\mu\sigma}\nabla^2 \mathbf{B}$$

. For $Rm << 1$ implies $\mu\sigma << 1$. In such case, the effective equation becomes

$$\frac{\partial \mathbf{B}}{\partial t} = \frac{1}{\mu\sigma}\nabla^2 \mathbf{B}$$

. This is the diffusion equation. It implies that field variations on a length scale L are destroyed over a diffusion time scale, τ_d. Hence, the induced magnetic field **b** is neglected, and from equation(2.1c) $e \sim Ub$, which is very small. Therefore, in equation (2.1c)

$$\nabla \times \mathbf{e} + \nabla \times \mathbf{E}_0 = -\frac{\partial(\mathbf{B}_0 + \mathbf{b})}{\partial t}$$

Where, $\mathbf{E}_0$ and $\mathbf{B}_0$ are external static electric and magnetic fields. after neglecting **b** and **e**

$$\nabla \times \mathbf{E}_0 \approx 0 \quad \text{or} \quad \mathbf{E} \approx 0$$

$$\therefore \mathbf{E} = -\nabla\phi$$

where ϕ is electric scalar potential

- In electromagnetic force, $\mathbf{f}_e$, the electric term is very small then the magnetic term. Hence we can neglect $\lambda\mathbf{E}$.

- We consider incompressible flow. Hence, the equation (2.1h) becomes

$$\nabla \cdot \mathbf{u} = 0$$

.

The MHD equations (2.1) are simplified as follows:

$$\nabla \cdot \mathbf{B} = 0 \tag{2.2a}$$

$$\nabla \times \mathbf{E} = 0 \tag{2.2b}$$

$$\nabla \times \mathbf{B} = \mu \mathbf{J} \tag{2.2c}$$

$$\mathbf{J} = \sigma \left(-\nabla \phi + \mathbf{u} \times \mathbf{B} \right) \tag{2.2d}$$

$$\nabla \cdot \mathbf{J} = 0 \tag{2.2e}$$

$$\frac{\partial \mathbf{u}}{\partial t} - 2\nu \nabla \cdot \mathbb{D}(\mathbf{u}) + \mathbf{u} \cdot \nabla \mathbf{u} + \frac{1}{\rho} \nabla p = \frac{1}{\rho} \mathbf{f_b} + \frac{1}{\rho} \mathbf{J} \times \mathbf{B} \tag{2.2f}$$

$$\nabla \cdot (\mathbf{u}) = 0 \tag{2.2g}$$

2.2 Assumptions behind SM82 model

As per Sommeria and Moreau, if the Stuart number and Hartmann number are very large in 3D magnetohydrodynamic flow, then the velocity components becomes invariant along the direction of magnetic field exhibiting 2D behavior, except in regions close to the walls known as Hartmann layer, where it still exhibits 3D behavior. These layers are situated along the insulated walls perpendicular to the magnetic field where viscous force produces strong velocity gradient. Thus, the complete flow is divided into two regions, (i). bulk flow where flow in magnetic field direction is invariant and (ii). flow in Hartmann layer. This nature of the flow is called quasi-2 Dimensional (Q2D) flow. Such flows with high Stuart number and high Hartmann number can be handled by Q2D approximation. The characteristic thickness of this Hartmann layer is usually a/Ha. Modeling of this layer requires very fine meshing and therefore is usually modeled by using some exponential profile. The study of Hartmann layer results in boundary condition for the bulk flow. This bulk flow experiences the braking effect produced due to viscous effect in Hartmann layer, which is incorporated in Navier-Stokes equation. There is another layer formed near walls parallel to the magnetic field, called Shercliff layer. Its characteristic thickness is usually $a/\sqrt{Ha}$. The layer usually is passive in character and do not react back upon the bulk flow. This layer also exhibits 3D behavior,

however, it is of little consequence on Q2D flow since the Q2D approximation involves a local error within 10% and the error is significant only inside the Shercliff layer. With these justifications, the model of Sommeria and Moreau is used which is derived by averaging the flow equations along the direction of magnetic field lines. This model is called SM82 model.
Along with the MHD flow equation, to study the effect of turbulent flow on heat transfer we also consider the Convection-Diffusion Equation, where the 3D temperature is also approximated to 2D temperature by averaging along the direction of magnetic field lines. As per the discussions the final equations are

$$\begin{aligned} \frac{\partial \bar{\mathbf{u}}}{\partial t} - 2\nu\nabla\cdot\mathbb{D}(\bar{\mathbf{u}}) + \bar{\mathbf{u}}\cdot\nabla\bar{\mathbf{u}} + \frac{1}{\rho}\nabla\bar{p} &= -\frac{2B}{a}\sqrt{\frac{\sigma\nu}{\rho}}\bar{\mathbf{u}} \\ \nabla\cdot(\bar{\mathbf{u}}) &= 0 \\ \rho c_p\frac{\partial\bar{T}}{\partial t} + \rho c_p\left(\bar{\mathbf{u}}\cdot\nabla\right)\bar{T} &= k\nabla^2\bar{T} + \sigma\mathbf{B}^2\bar{\mathbf{u}}^2 \end{aligned} \tag{2.3}$$

Here, c_p is specific heat capacity, k is the thermal conductivity. The average pressure $\bar{p}$, temperature $\bar{T}$ and velocity $\bar{\mathbf{u}}$ fields in z direction are obtained using the following relations:

$$\bar{\mathbf{u}} = \frac{1}{a}\int_{z=0}^{z=a}\mathbf{u}dz;\ \bar{p} = \frac{1}{a}\int_{z=0}^{z=a}pdz;\ \bar{T} = \frac{1}{a}\int_{z=0}^{z=a}Tdz \tag{2.4}$$

To obtain equation dimensionless, we scale the the model by using the characteristic length, time, temperature, pressure. For this model, characteristic length $L = d$ where d is the cylinder dimension, characteristic velocity $U = u_m$, where u_m is the maximum of velocity profile. Characteristic pressure is $P_c = \rho u_m^2$, characteristic time, $t_h = d/u_m$ and characteristic temperature $\Delta T = T_w - T_o$ where T_w is the temperature of the heated wall as shown in Fig. (2.1) and T_o is free stream temperature. We obtain following equation.

$$\frac{\partial\mathbf{u}}{\partial t} - \frac{2}{Re}\nabla\cdot\mathbb{D}(\mathbf{u}) + \mathbf{u}\cdot\nabla\mathbf{u} + \nabla p + \hat{\kappa}\mathbf{u} = 0 \tag{2.5}$$

$$\nabla\cdot\mathbf{u} = 0 \tag{2.6}$$

$$\frac{\partial\Theta}{\partial t} - \frac{1}{Pe}\Delta\Theta + (\mathbf{u}\cdot\nabla)\Theta = J_o\mathbf{u}^2 \tag{2.7}$$

$\mathbf{u}$, p and Θ are dimensionless quantities, $\hat{\kappa}$ is a scalar constant, P_e is the Peclet number and J_o is the joule heating. All of them are defined as,

$$\hat{\kappa} = 2\frac{d^2}{a^2}\frac{Ha}{Re} \quad , P_e = \frac{\mu c_p}{k}Re \quad \text{and} \quad J_o = \frac{\sigma B^2 u_m d}{\rho c_p \Delta T}$$

J_o is very small (of order of 10^{-5}) so the Joule heating term is neglected.

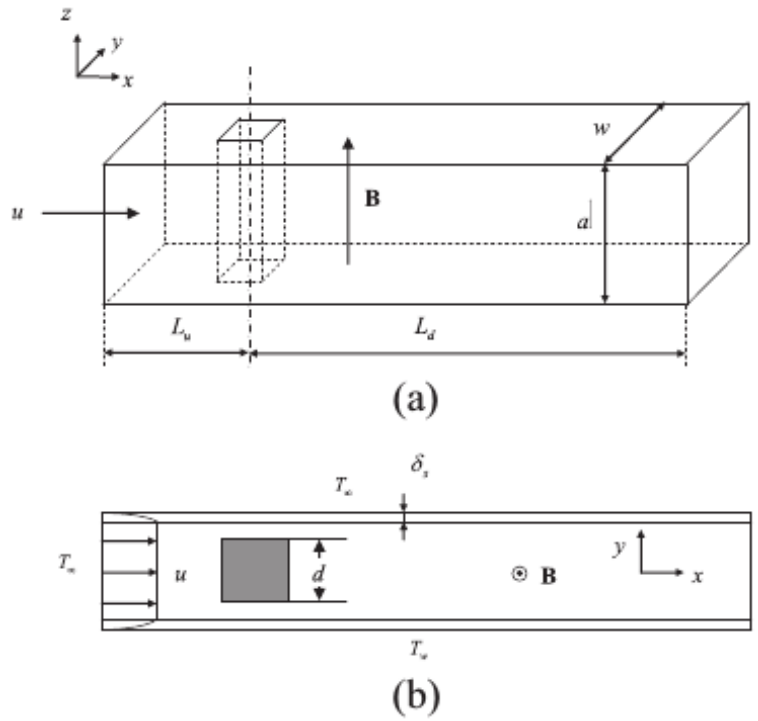

Figure 2.1: (a) Actual configuration of problem. (b) Equivalent 2D problem, taken from [3].

2.3 Related work

Several works on the investigation of MHD flows have been published in the literature. A few works have considered duct to be electrically conducting, others to be completely insulated. Many have studied the MHD flow in the presence of obstacles inside the duct of various shapes and sizes.

Sommeria and Moreau have proposed a quasi-static model famously known as the SM82 model in [6]. The authors have studied 2D dynamics observed in duct flow in terms of the effect of insulating walls. Further, They have discussed the conditions under which MHD turbulent

flow becomes 2D (or quasi 2D model). Moreover, They have mentioned that the quasi 2D model is valid for low magnetic Reynolds number Rm and large interaction parameter, N.

Muck et al. [15] have simulated 3D MHD flow for Reynolds number of 200 and 250, around an insulated square cylinder in an insulated square duct under the presence of external uniform magnetic field aligned with the cylinder. The authors have discussed the transition from initial 3D flow to quasi-2D flow. Further, the author computed Hartmann layer by applying the appropriate model. Finally, they have concluded that 3D flow transitions to quasi-2D flow for interaction parameter, N $\geq$ 1and confirmed the theory proposed by Sommeria and Moreau.

Dousset et al. [16] have simulated using finite volume method, the MHD liquid metal flow around a circular object in the presence of strong magnetic field aligned with the cylinder axis, for Re up to 6000 and Hartmann number, Ha in the range of 160 to 2160. To avoid expensive full 3D computation advantage of quasi-2D approximation has been exploited by neglecting velocity variations along the direction of the magnetic field for low Reynolds number. They have discussed various flow regimes and its effect due Ha and claim that first three flow regimes, i.e. creeping flow regimes **I**, the twin vortex regime **II** and the regular Karman vortex shedding regime **III** are similar to those of the non-magnetohydrodynamics flow with transitions controlled by Ha/Re, and major changes is visible in IV regime where Karman vortices becomes irregular. The effect of Ha on Karman Vortex Street, viscous drag forces and re-circulation length is also analyzed. An attempt has been made to find out whether the model is valid for high Reynolds number 3×10^4 and Ha 1120. The authors have concluded that the SM82 model may give only qualitative results rather than quantitative results and thus 3D simulation is inevitable.

Shahri et al. [17] and Dipankar et al. [3] have simulated the MHD liquid metal flow around a square object in the presence of strong magnetic field aligned with the cylinder axis, for Re up to 6000 and Hartmann number, Ha in the range of 160 to 2160. They have used finite volume based CFD solver Ansys Fluent. The main aim of this research was to study the effect of Ha on heat transfer. They have solved a convection-diffusion equation along with qusi-2D MHD equation and investigated the average Nusselt number against Ha.

Chapter 3

Turbulence Modelling

3.1 Turbulence

Turbulent flows are highly unsteady flows, characterized by chaotic behavior, where the main velocity flow field is superimposed by random velocity fluctuations. These random velocity fluctuations give rise to flow structures (eddies, scales) of different scale lengths or sizes. During the numerical solution of turbulent flows using, simulation of smallest eddies poses the greatest challenge as they are most difficult to resolve. The prevailed approaches for simulation of turbulent fluid flows are Direct-Numerical Solution (DNS), Reynolds-averaged Naiver-Stokes (RANS) and Large eddy simulation (LES). In DNS, Naiver-stokes equation (NSE) are numerically solved by resolving all scales by an appropriate fine mesh. In RANS approach the Naiver-stokes equation is time averaged into a steady and fluctuating part where the fluctuating part is then modeled by appropriate models. In LES the NSE is spatially averaged by some filter functions and the entire range of scales are segregated into resolved large and unresolved small scales. The larger scales of the flow are computed with the effect of small unresolved scales on the large resolved incorporated by a turbulence model. DNS can be computationally very expensive as very fine mesh is required for resolution of all scales in turbulence flow. In LES, the major challenges associated are (i) Choice of the appropriate filter functions, and (ii) The commutation errors which occur as the theory of LES developed for unbounded domain get restricted to the bounded domain.

3.2 Three scale Variational Multiscale method

The Variational Multiscale Method (VMS) is an improvement over classical LES approach for turbulence modeling and falls under the paradigm of scale separating schemes. The theoretical framework of VMS has initially been established in [7]. Initially, the VMS involved dividing the entire range of scales into two groups the "resolved scales" and the "unresolved scales", later it was extended to three scale separation, Collis [8] Gravemeir [13]. Un three scale separation the resolved scales was further subdivided into resolved small and resolved large scales; see Fig.(3.1). Apart from the simulation of turbulent flow, the multiscale phenomena have been applied to problems such as convection dominated Convection-diffusion problem in [12].

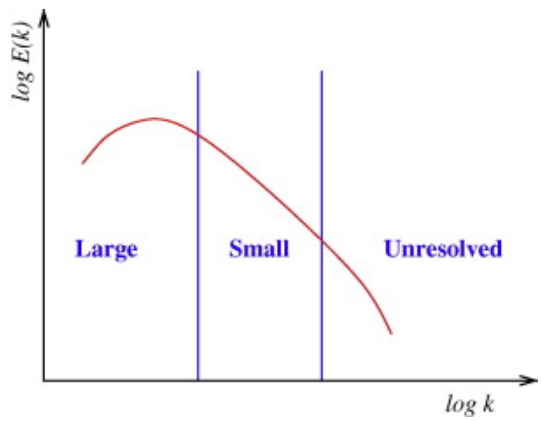

Figure 3.1: Schematic of the turbulent energy spectrum as a function of wavenumber, with scale partitioning for VMS. taken from [4]

To understand three scale VMS, consider a simple variational formulation of MHD equations (2.5) and (2.6).

$$A\left(\mathbf{u};(\mathbf{u},p),(\mathbf{v},q)\right)=\mathbf{F}(\mathbf{v}) \tag{3.1}$$

Where,
$$A\left(\mathbf{u};(\mathbf{u},p),(\mathbf{v},q)\right)=(\mathbf{u}_t,\mathbf{v})+(2\nu\mathbb{D}(\mathbf{u}),\mathbb{D}(\mathbf{v}))+(\mathbf{u}\cdot\nabla\mathbf{u},\mathbf{v})$$
$$-(p,\nabla\cdot\mathbf{v})+(q,\nabla\cdot\mathbf{u})+\hat{\kappa}\left(\mathbf{u},\mathbf{v}\right)$$

and,
$$\mathbf{F}(\mathbf{v})=(\mathbf{f},\mathbf{v})$$

As explained in [11], the flow is decomposed into three scales such that $\mathbf{u}=\overline{\mathbf{u}}+\widetilde{\mathbf{u}}+\widehat{\mathbf{u}}$ and

$p = \overline{p} + \widetilde{p} + \widehat{p}$. Where, $(\overline{\mathbf{u}}, \overline{p})$ represents large scales, $(\widetilde{\mathbf{u}}, \widetilde{p})$ represents resolved small scales and $(\widehat{\mathbf{u}}, \widehat{p})$ represents unresolved small scales. Now decomposing the test functions into three scales in equation (3.1), and replacing (**u**,p) with scales, we obtain following coupled equations

$$A\left(\mathbf{u}; (\overline{\mathbf{u}}, \overline{p}), (\overline{\mathbf{v}}, \overline{q})\right) + A\left(\mathbf{u}; (\widetilde{\mathbf{u}}, \widetilde{p}), (\overline{\mathbf{v}}, \overline{q})\right) + A\left(\mathbf{u}; (\widehat{\mathbf{u}}, \widehat{p}), (\overline{\mathbf{v}}, \overline{q})\right) = (\mathbf{f}, \overline{\mathbf{v}}) \tag{3.2}$$

$$A\left(\mathbf{u}; (\overline{\mathbf{u}}, \overline{p}), (\widetilde{\mathbf{v}}, \widetilde{q})\right) + A\left(\mathbf{u}; (\widetilde{\mathbf{u}}, \widetilde{p}), (\widetilde{\mathbf{v}}, \widetilde{q})\right) + A\left(\mathbf{u}; (\widehat{\mathbf{u}}, \widehat{p}), (\widetilde{\mathbf{v}}, \widetilde{q})\right) = (\mathbf{f}, \widetilde{\mathbf{v}}) \tag{3.3}$$

$$A\left(\mathbf{u}; (\overline{\mathbf{u}}, \overline{p}), (\widehat{\mathbf{v}}, \widehat{q})\right) + A\left(\mathbf{u}; (\widetilde{\mathbf{u}}, \widetilde{p}), (\widehat{\mathbf{v}}, \widehat{q})\right) + A\left(\mathbf{u}; (\widehat{\mathbf{u}}, \widehat{p}), (\widehat{\mathbf{v}}, \widehat{q})\right) = (\mathbf{f}, \widehat{\mathbf{v}}) \tag{3.4}$$

Here, the linearity the variational problem with respect to the test functions has been used. Now we shall be making the following assumptions for modifying the above equations,

- The equations with test function from the unresolved scales, i.e. Eq (3.4) is neglected as the unresolved scales are beyond the computational capabilities of our resource.
- It is assumed that the unresolved scales do not influence the large scales directly, i.e. the form $A\left(\mathbf{u}; (\widehat{\mathbf{u}}, \widehat{p}), (\overline{\mathbf{v}}, \overline{q})\right)$ can be done away with
- The influence of unresolved scales onto the small resolved scales is modeled:as $A\left(\mathbf{u}; (\widehat{\mathbf{u}}, \widehat{p}), (\widetilde{\mathbf{v}}, \widetilde{q})\right) \approx B\left(\mathbf{u}; (\overline{\mathbf{u}}, \overline{p}), (\widetilde{\mathbf{u}}, \widetilde{p}), (\widetilde{\mathbf{v}}, \widetilde{q})\right)$ or $B\left((\overline{\mathbf{u}}, \overline{p}), (\widetilde{\mathbf{u}}, \widetilde{p}), (\widetilde{\mathbf{v}}, \widetilde{q})\right)$.

Finally, Combining Equations(3.2) and (3.3) with the assumptions we obtain

$$A\left(\mathbf{u}_H; (\mathbf{u}_H, p_H), (\mathbf{v}_H, q_H)\right) + B\left((\overline{\mathbf{u}}, \overline{p}), (\widetilde{\mathbf{u}}, \widetilde{p}), (\widetilde{\mathbf{v}}, \widetilde{q})\right) = (\mathbf{f}, \mathbf{v}_H) \tag{3.5}$$

where, $(\mathbf{u}_H, p_H)$ represents the combination of large and small resolved scales. The choice of $B\left((\overline{\mathbf{u}}, \overline{p}), (\widetilde{\mathbf{u}}, \widetilde{p}), (\widetilde{\mathbf{v}}, \widetilde{q})\right)$ depends on turbulence model used. For current problem we consider the Smagorinsky eddy viscocity model. From physical point of view addition of turbulence model is infact addition of artificial viscosity into the system, specifically the Smagorinsky model is a diffusive model which adds stability to the system.

Finally, the important aspects of VMS are:

i All scales are separated based on projection into appropriate spaces, hence it allows for separation of the entire range of scales in the flow field (usually two or three scales), thus enabling different numerical treatment for different scales.

ii It avoids the difficulty of finding appropriate filter functions

iii Filter functions are used only for unresolved small scales thus reducing commuting error.

There are several variants of VMS proposed in literature; [7–14]. Out of them, two most commonly used approaches are (i) Residual-based VMS and (ii) Projection based VMS. Since projection based VMS is computationally more efficient than residual based with some compromise in accuracy of the solution, projection based VMS has been considered for the current problem.

3.3 Literature Review

The implementation of projection based variational multiscale finite element method for the incompressible Naiver-Stokes equations have been discussed in Volker John et al. [18]. The author has in detail explained the spaces considered for projection based variational formulation and system matrix construction.

Ramon et al. [19] presented a stabilized formulation using a residual-based variational multiscale (VMS) framework, proposed by [7], to solve the induction-less MHD problem using FEM. The authors have used a monolithic approach where the systems of partial differential equations consist of coupled Naiver-Stokes equation and Darcy-type problem via Lorentz's force and generated current terms.

Chapter 4

Mathematical Modelling

4.1 Mathematical Model

4.1.1 Governing Equations

The dimensionless modified MHD equations obtained after considering the assumption for the quasi-two-dimensional (Q2D) or SM82 model along with convection-diffusion equation are:

$$\frac{\partial \mathbf{u}}{\partial t} - 2\nu\nabla \cdot \mathbb{D}(\mathbf{u}) + \mathbf{u} \cdot \nabla\mathbf{u} + \nabla p + k\mathbf{u} = 0 \tag{4.1}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{4.2}$$

$$\frac{\partial \Theta}{\partial t} - \frac{1}{Pe}\Delta\Theta + (\mathbf{u} \cdot \nabla)\Theta = 0 \tag{4.3}$$

4.1.2 Boundary Condition

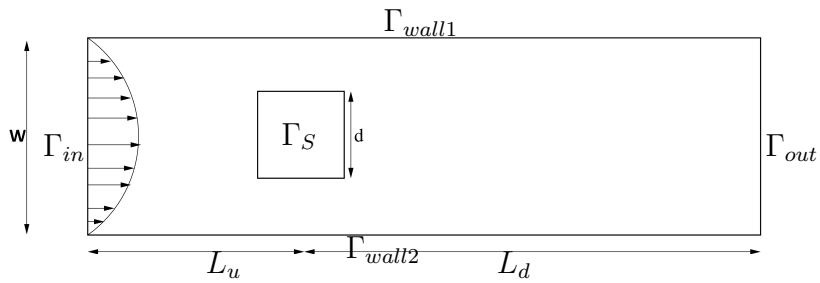

Figure 4.1: Computational domain for a fluid flow around an obstacle.

As per the schematic diagram depiction of the domain in Fig.(4.1), for the velocity field **u**, Neumann boundary condition is imposed at outflow Γ_{out} with $\frac{\partial \mathbf{u}}{\partial \mathbf{n}} = 0$, No-slip dirichlet boundary condition is imposed at Γ_{wall1}, Γ_{wall2}, and Γ_S with $\mathbf{u} = 0$, and at Γ_{in} the duct inflow is given by:

$$u(-L_u, y)^I = u_m \left[\frac{cosh(yd\frac{\sqrt{2Ha}}{a}) - cosh(h\frac{\sqrt{2Ha}}{2a})}{1 - cosh(h\frac{\sqrt{2Ha}}{2a})} \right]$$
$$u^{II} = 0$$

For Temperature, dirichlet boundary condition has been imposed with $\Theta = 0$, at Γ_{in}, Γ_{wall2} and $\Theta = 1$ at Γ_{wall1}. At Γ_{out} zero neumann boundary condition is imposed

4.2 Magnetohydrodynamics equation

4.2.1 Standard Galerkin weak Formulation

Now we try to obtain weak formulation for Eqs. (4.1) and (4.2). For this consider the functional spaces V and Q such that $V = H^1(\Omega)^2$ and $Q = L^2(\Omega)$. Now, Let **v** and q be the test functions for **u** and p respectively. The weak form is obtained by multiplying equations (4.1) and (4.2) corresponding test functions and integrating over the domain Ω with second order terms being integrated by part. We obtain the weak form which read as $\mathbf{u} : [0, T] \to V, p : (0, T] \to Q$

$$\begin{aligned} (\mathbf{u}_t, \mathbf{v}) + (2\nu\mathbb{D}(\mathbf{u}), \mathbb{D}(\mathbf{v})) + (\mathbf{u} \cdot \nabla\mathbf{u}, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) + \hat{\kappa}\,(\mathbf{u}, \mathbf{v}) = 0 \;\; \forall \mathbf{v} \in V \\ (q, \nabla \cdot \mathbf{u}) = 0 \;\; \forall q \in Q \end{aligned} \tag{4.4}$$

4.2.2 Weak formulation with variational multiscale stabilized scheme

For current problem projection projection based variational multiscale method is used. First we consider the triangulation of the domain Ω into simplex cells represented by Ω_h and further $V_h(\Omega_h) \subset V$ and $Q_h(\Omega_h) \subset Q$ be a pair of conforming finite element spaces satisfying the

discrete inf-sup condition which is descibed as follows:

$$\inf_{q_h \in Q_h} \sup_{\mathbf{v}_h \in V_h} \frac{(q_h, \nabla \cdot \mathbf{v}_h)}{\|q_h\|_{L^2} \; \|\nabla \mathbf{v}_h\|_{L^2}} \geq \beta \geq 0 \tag{4.5}$$

where β is some positive constant, independent of the mesh size h. It is to be noted that V and Q are the continuous spaces which contain the entire range of scales, whereas V_h and Q_h are finite dimensional and they contain only the resolved large- and small-scales. Moreover, let us consider a space of 2×2 symmetric tensors (deformation tensor in the context of NSE) as follows:

$$L = \left\{ \mathbb{L} : \mathbb{L} \in (L^2(\Omega))^{2\times 2}, \, \mathbb{L} = \mathbb{L}^T \right\}$$

and further we consider $L_H \subset L$ to be a coarse discrete space defined as

$$L_H = \left\{ \mathbb{D}(\mathbf{v}) : \mathbb{D}(\mathbf{v}) \in L, \, \forall \, \mathbf{v} \in V_H \subset (H^1(\Omega))^2 \right\}.$$

Here, in the notations the subscript H in V_H has been used to denote that the polynomial order of the finite element space V_H is smaller than the polynomial order of the space V_h. Next, we shall define $P_{L_H} : L \to L_H$, $\mathbb{D}(\mathbf{v}) \mapsto P_{L_H}\mathbb{D}(\mathbf{v})$ using the L^2-projection operator

$$((\mathbb{I} - P_{L_H})\mathbb{D}(\mathbf{u}), \, \mathbb{L}_H) = 0, \quad \forall \; \mathbb{D}(\mathbf{u}) \in \; L, \; \; \mathbb{L}_H \in L_H.$$

Making use of these defined spaces and the projection operator, the (continuous in time) semi-discrete VMS form of the MHD equations reads as : For given $\mathbf{u}_0$, find $\mathbf{u}_h \in V_h$, $p_h \in Q_h$ and $\mathbb{G}_H = P_{L_H}\mathbb{D}(\mathbf{u}_h) \in L^2(0, \mathrm{T}, L_H)$ such that for all $(\mathbf{v}_h, q_h) \in V_h \times Q_h$ and $\mathbb{L}_H \in L_H$.

$$(\mathbf{u}_{ht}, \mathbf{v}_h) + (\tilde{\nu}\mathbb{D}(\mathbf{u}_h), \mathbb{D}(\mathbf{v}_h)) + (\mathbf{u}_h \cdot \nabla \mathbf{u}_h, \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) +$$
$$\hat{\kappa}\,(\mathbf{u}_h, \mathbf{v}_h) - (\nu_T \mathbb{G}_H, \mathbb{D}(\mathbf{v}_h)) = 0 \quad \forall \quad \mathbf{v}_h \in V_h \tag{4.6a}$$

$$(q_h, \nabla \cdot \mathbf{u}_h) = 0 \quad \forall \quad q_h \in Q_h \tag{4.6b}$$

$$(\mathbb{G}_H - \mathbb{D}(\mathbf{u}_h), \mathbb{L}_H) = 0 \quad \forall \quad \mathbb{L}_H \in L_H \tag{4.6c}$$

Here, $\tilde{\nu} = 2\nu + \nu_T$, where $\nu_T \geq 0$ is turbulent viscosity model.

Remark 1: In this scheme, care must be taken in suitably choosing the space L_H and the turbulence model ν_T, and these two choices characterize the modeling of the subgrid scales in this projection based VMS scheme. The first equation in (4.6) represents the turbulent viscosity to be added to all resolved scales and then subtracted from only the resolved large scales. By this manipulation it's made sure that the effect of turbulent viscosity model is limited to only the resolved small scales.

Remark 2: The L^2-projection of $\mathbb{D}(\mathbf{u}_h)$ from L onto L_H is represented in the third equation in (4.6) Here, the space L can be interpreted as the deformation tensor of the resolved scales, with both the large and the small scales combined, whereas the deformation tensor of the resolved large scales. has been represented by L_H.

Now, let V_h , Q_h , and L_H be spanned with basis functions θ , ψ, and , l such that

$$V_h = \text{span}\left\{\theta_i^I = \begin{pmatrix}\theta_i \\ 0\end{pmatrix}, \theta_i^{II} = \begin{pmatrix}0 \\ \theta_i\end{pmatrix} : i = 1, 2, ..., N\right\}$$

$$Q_h = \text{span}\left\{\psi_l : l = 1, 2, ..., M\right\} \text{ and}$$

$$L_H = \text{span}\left(l_j^1 = \begin{Bmatrix} l_{Hj} & 0 \\ 0 & 0 \end{Bmatrix}; l_j^2 = \frac{1}{2}\begin{Bmatrix} 0 & l_{Hj} \\ l_{Hj} & 0 \end{Bmatrix}; l_j^3 = \begin{Bmatrix} 0 & 0 \\ 0 & l_{Hj} \end{Bmatrix} : i = 1, 2..., N\right)$$

then,

$$\mathbf{u}_h = \sum_{j=1}^{N} u_j^I \theta_j^I(x) + u_j^{II}\theta_j^{II}(x) \qquad p_h = \sum_{l=1}^{M} p_l \psi_l(x)$$

$$\mathbb{G}_H = \sum_{j=1}^{N} g_j^{11} l_j^1 + \sum_{j=1}^{N} g_j^{12} l_j^2 + \sum_{j=1}^{N} g_j^{22} l_j^3$$

4.2.3 Fully Discrete Form

We have used Crank-Nicolson method for time discretization in both Eqs. (4.1). Let $0 \leq t_0 < t_1 ..... < t_n =$ T be the decomposition of the time interval with uniform time steps, $\delta_t^n = t^{n+1} - t^n$.

$$\begin{aligned}\left(\frac{\mathbf{u}_h^n - \mathbf{u}_h^{(n-1)}}{\delta_t^n}, \mathbf{v}_h\right) &+ \frac{1}{2}\left(\tilde{\nu}\mathbb{D}(\mathbf{u}_h^n), \mathbb{D}(\mathbf{v}_h)\right) + \frac{1}{2}\left(\mathbf{u}_h^n \cdot \nabla \mathbf{u}_h^n, \mathbf{v}_h\right) \\ &+ \frac{1}{2}\hat{\kappa}\left(\mathbf{u}_h^n, \mathbf{v}_h\right) + \frac{1}{2}\left(\nu_T \mathbb{G}_H^n, \mathbb{D}(\mathbf{v}_h)\right) - \left(p_h, \nabla \cdot \mathbf{v}_h\right) \\ &= -\frac{1}{2}\left(\tilde{\nu}\mathbb{D}(\mathbf{u}_h^{n-1}), \mathbb{D}(\mathbf{v}_h)\right) - \frac{1}{2}\left(\mathbf{u}_h^{n-1} \cdot \nabla \mathbf{u}_h^{n-1}, \mathbf{v}_h\right) \\ &- \frac{1}{2}\hat{\kappa}\left(\mathbf{u}_h^{n-1}, \mathbf{v}_h\right) - \frac{1}{2}\left(\nu_T \mathbb{G}_H^{n-1}, \mathbb{D}(\mathbf{v}_h)\right) \qquad (4.7)\end{aligned}$$

$$\left(q_h, \nabla \cdot \mathbf{u}_h^n\right) = 0 \qquad (4.8)$$

The equation (4.6) is non-linear. To solve this, we use fixed point iteration method. After substituting $\mathbf{u}_h,\ p_h$ and $\mathbb{G}_H$ in Eqs. (4.6) and (4.7) we obtain the following linear algebraic equation

$$\left[M + \frac{\delta_t^n}{2}X\right]U^n = \left[M - \frac{\delta_t^n}{2}X\right]U^{n-1} \qquad (4.9)$$

Where,

$$X = \begin{bmatrix} A_{11} & A_{12} & B_1^T & \tilde{G_{11}} & \tilde{G_{12}} & \tilde{G_{13}} \\ A_{21} & A_{22} & B_2^T & \tilde{G_{21}} & \tilde{G_{22}} & \tilde{G_{23}} \\ B_1 & B_2 & 0 & 0 & 0 & 0 \\ G_{11} & G_{12} & 0 & N & 0 & 0 \\ G_{21} & G_{22} & 0 & 0 & N/2 & 0 \\ G_{31} & G_{32} & 0 & 0 & 0 & N \end{bmatrix};\ M = \begin{bmatrix} M_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & M_{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix};\ U = \begin{bmatrix} U^I \\ U^{II} \\ P \\ g^{11} \\ g^{12} \\ g^{22} \end{bmatrix}$$

$$[A_{11}]_{ij} = \tilde{\nu}\int_\Omega \left(\frac{\partial\theta_j^I}{\partial x}\frac{\partial\theta_i^I}{\partial x} + \frac{1}{2}\frac{\partial\theta_j^I}{\partial y}\frac{\partial\theta_i^I}{\partial y}\right) + \int_\Omega \left(u_j^I\frac{\partial\theta_j^I}{\partial x} + u_j^{II}\frac{\partial\theta_j^I}{\partial y}\right)\theta_i^I + \hat{\kappa}\int_\Omega \theta_j^I\theta_i^I$$

$$[A_{22}]_{ij} = \tilde{\nu}\int_\Omega \left(\frac{1}{2}\frac{\partial\theta_j^{II}}{\partial x}\frac{\partial\theta_i^{II}}{\partial x} + \frac{\partial\theta_j^{II}}{\partial y}\frac{\partial\theta_i^{II}}{\partial y}\right) + \int_\Omega \left(u_j^I\frac{\partial\theta_j^{II}}{\partial x} + u_j^{II}\frac{\partial\theta_j^{II}}{\partial y}\right)\theta_i^{II} + \hat{\kappa}\int_\Omega \theta_j^{II}\theta_i^{II}$$

$$[A_{12}]_{ij} = \tilde{\nu}\int_\Omega \left(\frac{1}{2}\frac{\partial\theta_j^{II}}{\partial x}\frac{\partial\theta_i^I}{\partial y}\right)$$

$$[A_{21}]_{ij} = \tilde{\nu}\int_\Omega \left(\frac{1}{2}\frac{\partial\theta_j^I}{\partial y}\frac{\partial\theta_i^{II}}{\partial x}\right)$$

$$[G_{11}]_{ij} = \int_\Omega l_i^H\frac{\partial\theta_j^I}{\partial x} \qquad [G_{21}]_{ij} = \int_\Omega \frac{1}{2}l_i^H\frac{\partial\theta_j^I}{\partial y} \qquad [G_{31}]_{ij} = 0$$

$$[G_{12}]_{ij} = 0 \qquad [G_{22}]_{ij} = \int_\Omega \frac{1}{2}l_i^H\frac{\partial\theta_j^{II}}{\partial x} \qquad [G_{32}]_{ij} = \int_\Omega l_i^H\frac{\partial\theta_j^{II}}{\partial y}$$

$$[\tilde{G_{11}}]_{ij} = \nu_T[G_{11}]_{ij}^T \qquad [\tilde{G_{12}}]_{ij} = \nu_T[G_{21}]_{ij}^T \qquad [\tilde{G_{13}}]_{ij} = 0$$

$$[\tilde{G_{21}}]_{ij} = 0 \qquad [\tilde{G_{22}}]_{ij} = \nu_T[G_{22}]_{ij}^T \qquad [\tilde{G_{23}}]_{ij} = \nu_T[G_{32}]_{ij}^T;$$

$$[N]_{ij} = \int_\Omega l_i^H l_j^H \qquad [B_1]_{il} = \int_\Omega \psi_l\frac{\partial\theta_i^I}{\partial x} \qquad [B_2]_{il} = \int_\Omega \psi_l\frac{\partial\theta_i^{II}}{\partial y}$$

$$\left[U^I\right] = \left(U_1^I, \ldots\ldots, U_N^I\right) \qquad \left[U^{II}\right] = \left(U_1^{II}, \ldots\ldots, U_N^{II}\right) \qquad [P] = (P_1, \ldots\ldots, P_M)$$

$$\left[g^{11}\right] = \left(g_1^{11}, \ldots\ldots, g_N^{11}\right) \qquad \left[g^{12}\right] = \left(g_1^{12}, \ldots\ldots, g_N^{12}\right) \qquad \left[g^{22}\right] = \left(g_1^{22}, \ldots\ldots, g_N^{22}\right)$$

Now, g^{11}, g^{12} and g^{22} can be solved using a saddle point method which leads to the linear algebra equation (4.8) for $(\mathbf{u}^h, p^h)$ change into

$$\left[M + \frac{\delta_t^n}{2}\tilde{X}\right]\tilde{U}^n = \left[M - \frac{\delta_t^n}{2}\tilde{X}\right]\tilde{U}^{n-1} \tag{4.10}$$

where,

$$\tilde{X} = \begin{bmatrix} \tilde{A_{11}} & \tilde{A_{12}} & B_1^T \\ \tilde{A_{21}} & \tilde{A_{22}} & B_2^T \\ B_1 & B_2 & 0 \end{bmatrix}; \ \tilde{U} = \begin{bmatrix} U^I \\ U^{II} \\ P \end{bmatrix}$$

$$\tilde{A_{11}} = A_{11} - G_{11}N^{-1}G_{11} - 2G_{12}N^{-1}G_{21} - G_{13}N^{-1}G_{31} \tag{4.11}$$

$$\tilde{A_{12}} = A_{12} - G_{11}N^{-1}G_{12} - 2G_{12}N^{-1}G_{22} - G_{13}N^{-1}G_{32} \tag{4.12}$$

$$\tilde{A_{21}} = A_{21} - G_{21}N^{-1}G_{11} - 2G_{22}N^{-1}G_{21} - G_{23}N^{-1}G_{31} \tag{4.13}$$

$$\tilde{A_{22}} = A_{22} - G_{21}N^{-1}G_{12} - 2G_{22}N^{-1}G_{22} - G_{23}N^{-1}G_{32} \tag{4.14}$$

4.3 Convection-Diffusion equation formulation

4.3.1 Weak Formulation

Consider function space V_Θ, such that $V_\Theta = H^1(\Omega)$. Let Φ be the test function for Θ. The weak form is obtained by multiplying equation (4.3) by the test functions Φ and integrating over the domain Ω with second order terms being integrated by part, which reads as $\Theta : [0, T] \to V_\Theta$

$$\left(\frac{\partial \Theta}{\partial t}, \Phi\right) + \frac{1}{Pe}\left(\nabla\Theta, \nabla\Phi\right) + \left(\mathbf{u} \cdot \nabla\Theta, \Phi\right) = 0 \quad \forall \quad \Phi \in V_\Theta \tag{4.15}$$

4.3.2 Fully Discrete Form

For spatial discretization of equation (4.15), we considered finite dimensional discrete space $V_{\Theta_h} \subset V_\Theta$ spanned by the scalar basis functions ω then

$$\Theta_h = \sum_{j=1}^{N} \Theta_j \omega_j(x) \qquad \forall \Theta_h \in V_{\Theta_h}$$

Using Crank-Nicolson method for time discretization we obtain the following fully discrete form of equation (4.15)

$$\begin{aligned}\left(\frac{\Theta_h^n - \Theta_h^{n-1}}{\delta_t^n}, \omega\right) + \frac{1}{2Pe}\left(\nabla\Theta_h^n, \nabla\omega\right) + \frac{1}{2}\left(\mathbf{u}^n \cdot \nabla\Theta_h^n, \omega\right) = &- \frac{1}{2Pe}\left(\nabla\Theta_h^{n-1}, \nabla\omega\right) \\ &- \frac{1}{2}\left(\mathbf{u}^n \cdot \nabla\Theta_h^{n-1}, \omega\right)\end{aligned} \tag{4.16}$$

Finally, we get the following system of algebraic equation from equation (4.16)

$$\left[A + \frac{\delta_t^n}{2} C^n\right] \Theta^n = \left[A - \frac{\delta_t^{n-1}}{2} C^{n-1}\right] \Theta^{n-1} \tag{4.17}$$

Where,

$$[A_{ij}] = \int_\Omega \omega_j \omega_i; \qquad [C_{ij}^n] = \int_\Omega \frac{1}{Pe}(\nabla\omega_j, \nabla\omega_i) + \int_\Omega (\mathbf{u}^n \cdot \nabla\omega_j, \omega_i)$$

and $\quad \{\Theta_i\} = \{\Theta_1, ..., \Theta_N\}$

4.4 Algorithmic implementation

The matrix equation (4.9) and (4.16) are solved in one-way coupled manner as shown in the algorithm

Algorithm 1 Solving System matrices

for $n = 1$ *to* t_max **do**

  for $k = 1$ *to* max_iter **do**

    $\left[M + \frac{\delta_t^n}{2}\tilde{X}\right] \tilde{U}^n = \left[M - \frac{\delta_t^n}{2}\tilde{X}\right] \tilde{U}^{n-1}$

  end

  $\left[A + \frac{\delta_t^n}{2} C^n\right] \Theta^n = \left[A - \frac{\delta_t^{n-1}}{2} C^{n-1}\right] \Theta^{n-1}$

end

Chapter 5

Numerical Results

The theoretical aspects of a projection based VMS scheme for MHD equations that we have discussed in detail in the previous sections have been validated by a comprehensive set of numerical examples in this section. In our computations we have used the inf-sup stable pair of finite element P_2/P_1 for the velocity and pressure, and additionally the space P_0 have been considered for large-scale velocity space. Further, the space P_2 has been considered as the ansatz and test space for temperature. For the subgrid scale model we have used the Smagorinsky eddy viscosity model which is defined as $\nu_T = c_t\delta^2||\mathbb{D}(u_h)||_F$, with $c_t = 0.01$, $\delta_h = 2h$ where h is the mesh size and $||\cdot||_F$ denotes the Frobenius norm. Further, all the computations have been carried out using ParMooN [20], an in house finite element of our research group.

5.1 Mesh Convergence

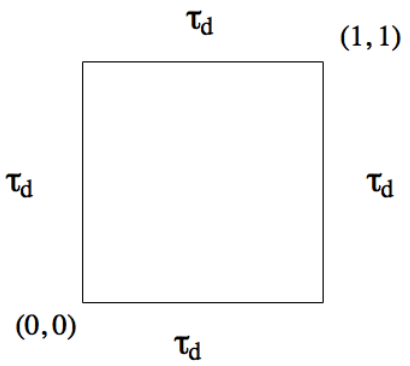

Figure 5.1: Unit square domain for mesh convergence

In this section, a mesh convergence study for the proposed numerical scheme is given. For this, a unit square domain; as shown in Fig.(5.1) with Dirichlet boundary condition τ_d at all the boundary for u^I, u^{II}, and Θ is considered . Following exact solutions are considered.

$$u^I = \sin(\pi x)\exp(-.1t), \qquad u^{II} = -\pi y\cos(\pi x)\exp(-.1t)$$

$$p = \sin(\pi x)\cos(\pi y)\exp(-.1t)$$

$$\text{and} \qquad \theta = \sin(2\pi x)\cos(2\pi y)\exp(-.1t)$$

The results are calculated for Ha =320, Re = 100, Pe = .02, and $\delta t = h/10$, where δt is time step and h is the mesh size. The plots for $\|err_u\|_{L^2(\Omega)}$, $|err_u|_{H^1(\Omega)}$ $\|err_p\|_{L^2(\Omega)}$, and $|err_p|_{H^1(\Omega)}$ $\|err_T\|_{L^2(\Omega)}$ $|err_T|_{H^1(\Omega)}$ is shown in figures (5.2), (5.3), and (5.4). err represents the error with the exact solution. The order of convergence obtained for velocity is 3, for pressure it is 2 and for temperature it is 3.

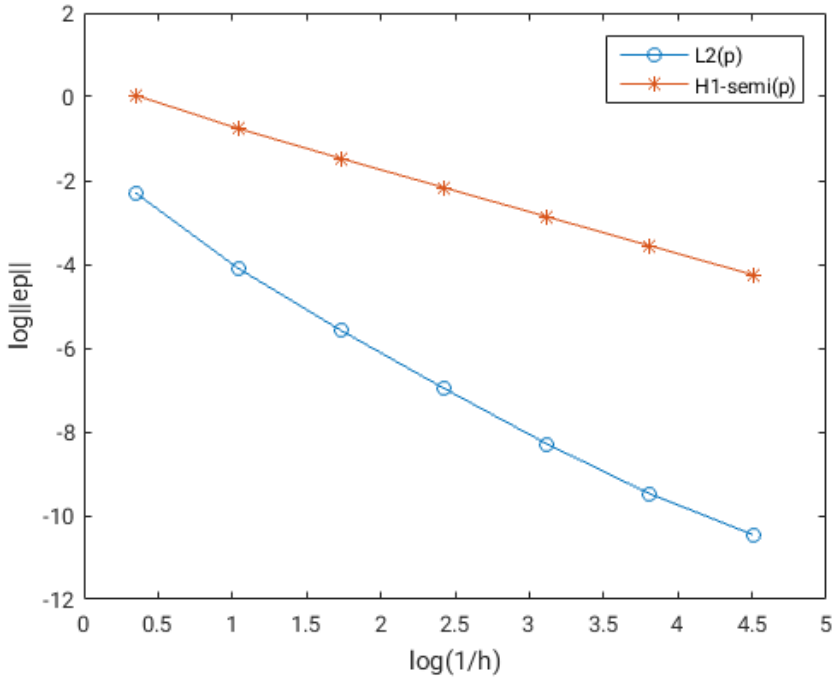

Figure 5.2: Convergence of pressure, Re=100

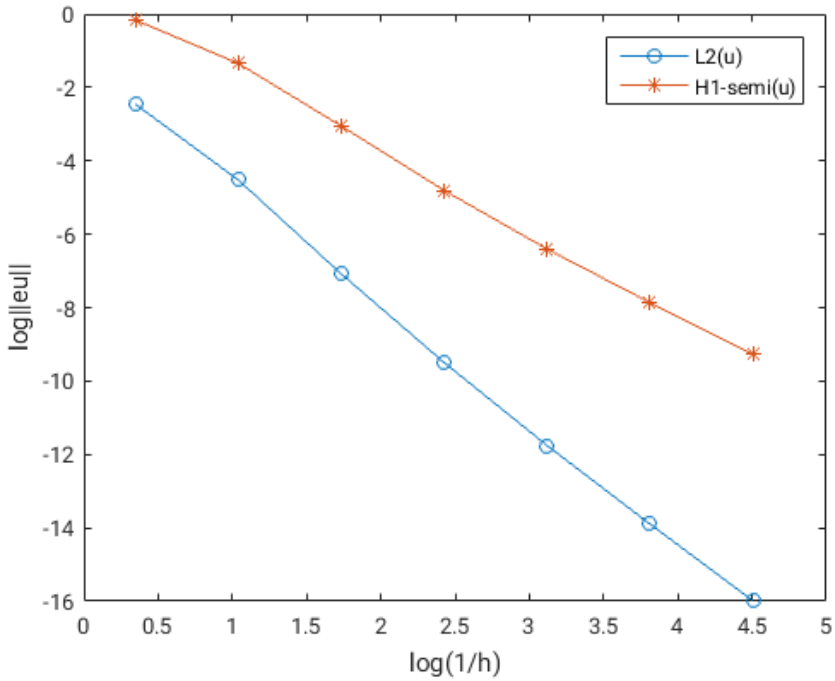

Figure 5.3: Convergence of Velocity, Re=100

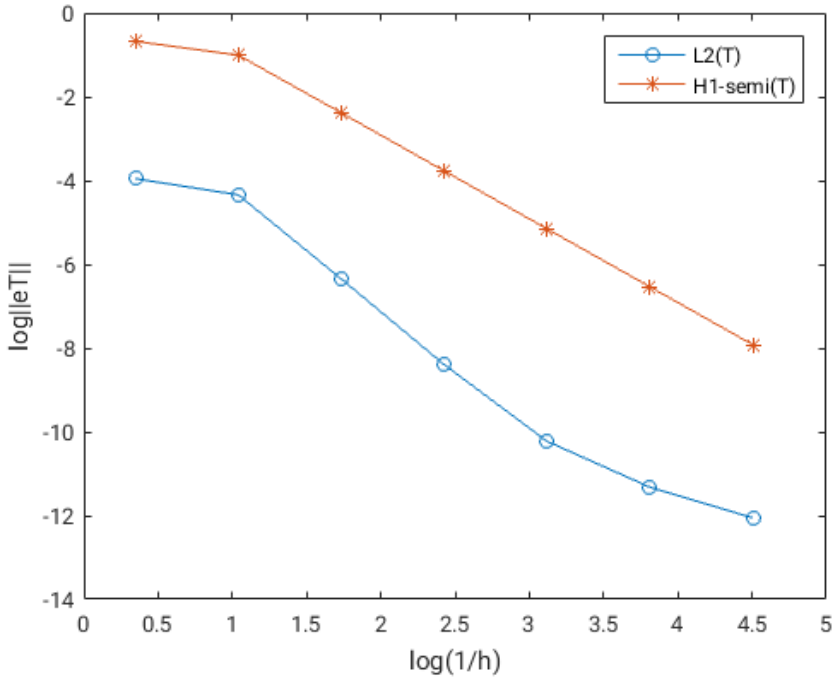

Figure 5.4: Convergence of Temperature, Re=100

5.2 Numerical example

5.2.1 Domain setup

A rectangular 2D duct of dimension $L \times w$ with an square obstacle of dimension d is considered. The width w is d/β and length L is $L_u + L_d$, where d is the dimension of the block. β is the blockage ratio, L_u is the distance from the inflow boundary to the centre of the obstacle and L_d is the distance of outflow boundary to centre of obstacle as shown in figure (2.1). For $\beta = .25$ triangle mesh is generated; Fig(5.5) using open source library TetGen [21].

5.2.2 Grid sensitivity test

In this results of grid sensitive analysis is discussed. The model is simulated for $t = 10t_h$ at $Re = 100$, $Ha = 320$ for different mesh size M1, M2, M3, M4 and M5. For each case,

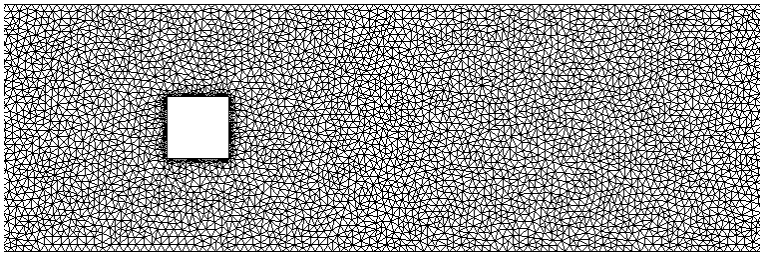

Figure 5.5: Generated Mesh

coefficient of drag due to viscous Cd_v, coefficient of lift Cd_p, using the following definition

$$C_D = \frac{2F_D}{\rho u_m^2 2d}$$

Similarly, space-time average Nusselt number $Avg_N u$ is also computed, definition of which is given in section (5.2.4). Next we, obtain the relative errors ε_{AvgNu}, ε_{Cd_v}, ε_{Cd_p} defined as

$$\varepsilon_{AvgNu} = \left|1 - \frac{AvgNu(Mi)}{AvgNu(M5)}\right|$$
$$\varepsilon_{Cd_v} = \left|1 - \frac{cl_v(Mi)}{cl_v(M5)}\right|$$
$$\varepsilon_{Cd_p} = \left|1 - \frac{cl_p(Mi)}{cl_p(M5)}\right|$$

The results obtained for different mesh size are shown in table (5.1). The relative error are observed to decrease with more fine mesh. Finally, type (M3) mesh is preferred keeping in view the accuracy of the results and computational convenience of the simulation.

Table 5.1: Grid Sensitivity analysis

Meshes	M1	M2	M3	M4	M5
Number of Cells	9808	15532	34652	68953	136176
Total Dofs	67686	104892	229172	453980	891864
ε_{AvgNu}	0.020253	0.003065	0.000230	0.0000172	0
ε_{Cd_v}	0.008827	0.005283	0.002520	0.0012880	0
ε_{Cd_p}	0.007850	0.009554	0.016000	0.0044578	0

5.2.3 Comparison of VMS stabilized and standard galerkin

The MHD equation used to study the behavior of the quasi-2D model is similar to the simple Navier-stokes equation with an extra reaction term. A stability analysis is done by simulating the flow using both simple Navier-stokes equation and the quasi-2D MHD equation for Reynolds number 5000, 7500, and 10000 without the implementation of VMS scheme, Fig.5.6. In each case, a number of time-step for which the simulation remains stable is obtained by setting the stopping condition as $residual > 1$. Here the $residual$ is defined as

$$residual = ||F - \hat{X}U||_{L_2(\Omega)}$$

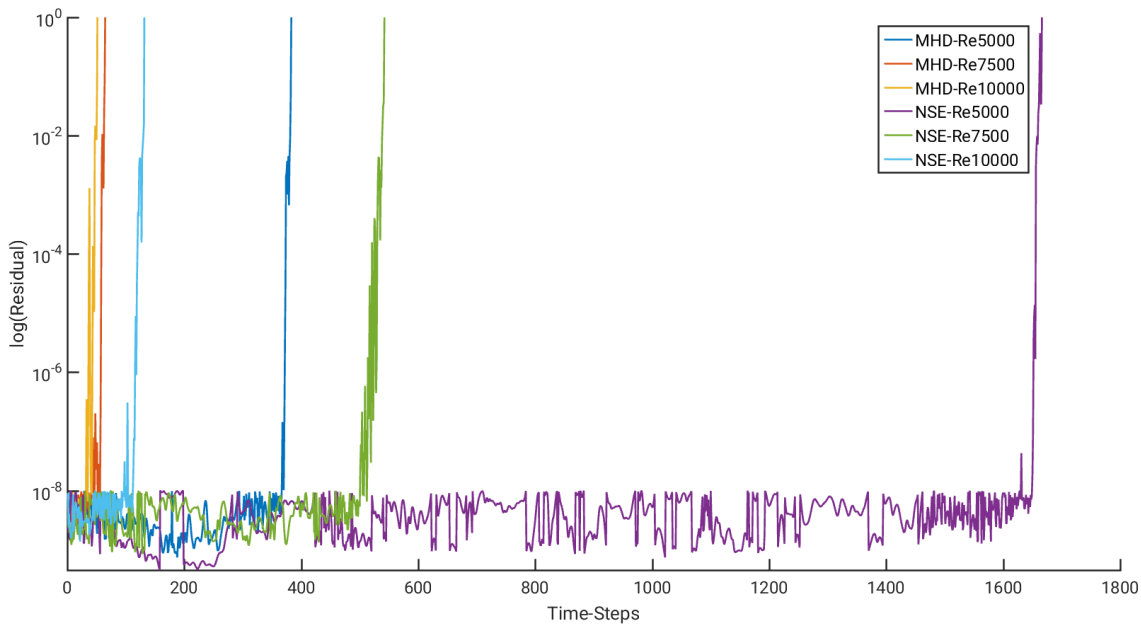

Figure 5.6: Comparison of NSE and MHD stability without VMS

As visible in the plots, the Navier-Stokes equation is more stable than MHD equation for all Reynolds number. This is due to fact that the extra reaction term, $k\mathbf{u}$, in the MHD equation is creating instability. Hence, there is also a need for some stabilization technique for the reaction term. Another stability analysis is done for the MHD equation with VMS scheme. As visible in Fig.5.7 the number of time-steps for which the flow is stable has increased. Hence, the VMS is capable of stabilizing the flow for high Reynolds number as well as stabilizing the

extra reaction term.

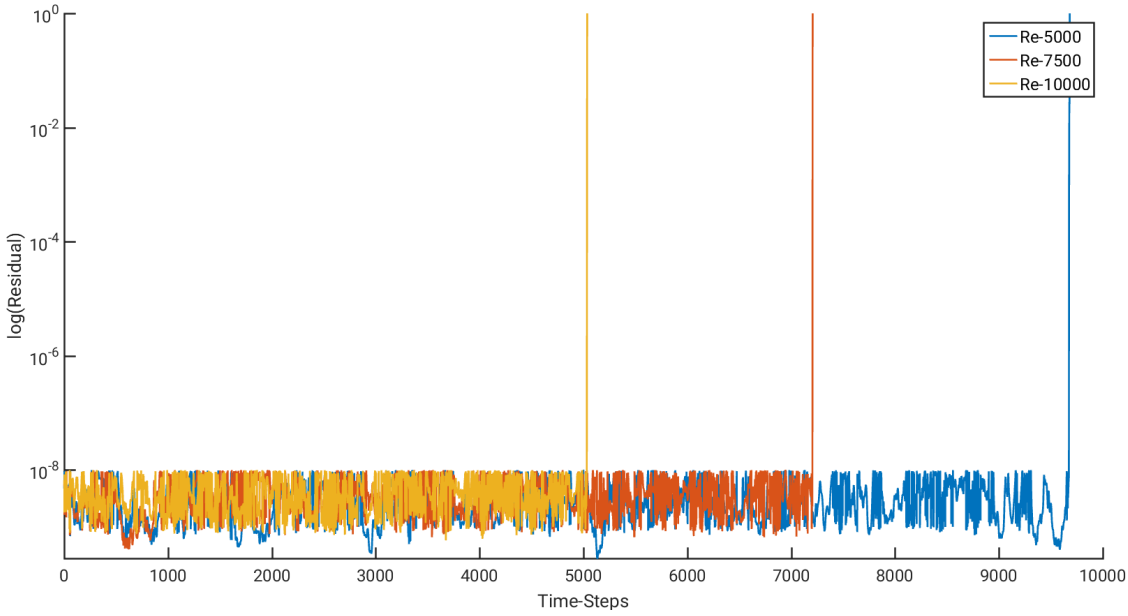

Figure 5.7: Plot of MHD stability with VMS

5.2.4 Nusselt number calculations

The average Nusselt number for different combination Reynolds number and Hartmann number at the heated bottom channel is calculated. The Nusselt number Nu is defined as the ratio of heat transfer via convection to conduction. Time average Nu is calculated for $t = 10t_h$. Time-surface-averaged Nusselt number is calculated as by first taking the time average of the local Nusselt number $(\overline{Nu}_w)$ at each x location along the heated wall and then integrating over the wall length L_t as shown in below equations.Here, T_f is the bulk fluid temperature.

$$Nu_w(x,t) = \frac{d}{T_f - T_w}\frac{\partial \Theta}{\partial y}\,|_{wall} \tag{5.1}$$

$$T_f = \frac{\int_0^w u\Theta dy}{\int_0^w u dy} \tag{5.2}$$

$$Nu = \frac{1}{\mathrm{L}_t}\int_0^{\mathrm{L}_t} \overline{Nu}_w dx \tag{5.3}$$

The average Nusselt number is calculated for Hartmann number 320, 1120 and 2160 and

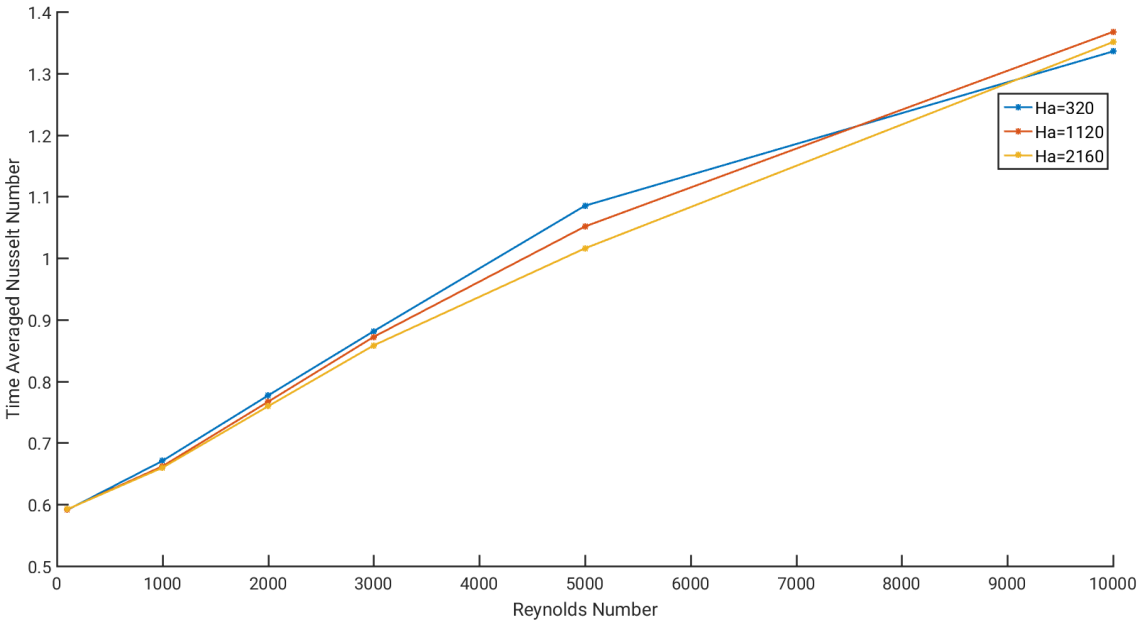

Figure 5.8: Plot of Nusselt number v/s Reynolds number

Re 100 to 10000. As visible in the in Fig.5.8 Nusselt number increases with the increase of Reynolds number. This shows that heat transfer through the convection increase with Reynolds number. It is also observed that increasing the Hartmann number will decrease the average Nusselt number. This is due to the fact that the increase in Hartmann number means external magnetic field strength is increased, hence it reduces the turbulence ultimately leading to reduced heat transfer. However, at around Re 9000, this phenomenon is not obeyed. This could be possible because the model's behavior is no more 2D. In such case, 3D simulation becomes inevitable to verify the results.

Fig.5.9 shows the plot of surface average Nusselt number with respect to time steps. It is observed that for lower Reynolds number, e.g. $Re = 100$ the heat transfer rate becomes constant with initial fluctuations whereas for higher Reynolds number the heat transfer does not become stable. This can be due to the interaction of secondary vortices generated from the walls with the Karman vortices originated from the block, which produces energy. With time flow with Re 10,000 creates more and more vortices and thus the calculated heat transfer is never stable.

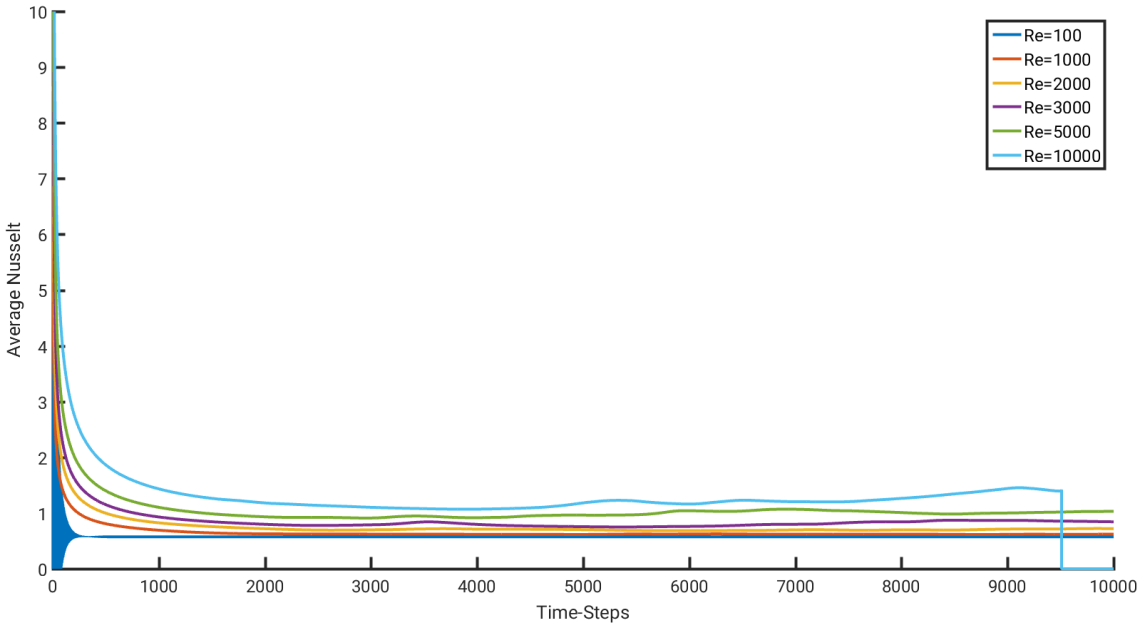

Figure 5.9: Plot of Nusselt number v/s Time Steps for Reynolds number 100 & 10000

5.2.5 Visualization

Fig.(5.10), shows the temperature in the domain for $Re = 1000$. As the Hartmann number is increased, the flow laminarizes and less fluctuation in temperature can be seen.

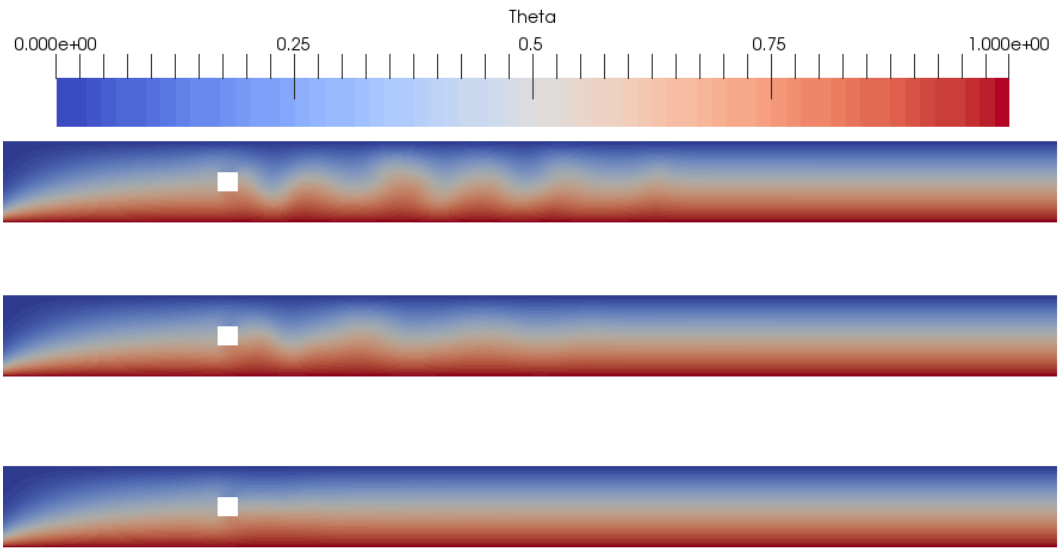

Figure 5.10: Temperature profile at Re = 1000 at (i) Ha = 320 (top), (ii) Ha = 1120 (middle), and (iii) Ha = 2160 (bottom)

Chapter 6

Conclusion

The thermo-fluidic transport of liquid metal around a square object placed in a square duct under a strong homogeneous magnetic field aligned with the cylinder axis is numerically simulated using a quasi-2D model. A wide range of Reynolds and Hartmann numbers are considered to study the behavior of the flow. In order to understand the thermal transport, the Nusselt number is calculated at the bottom heated channel wall. Apart from the Karman and secondary vortices observed during the simulation the other major observations are:

- VMS successfully simulated the flow with high Reynolds number 10,000.
- Variational multiscale method is able to simulate the turbulent flow as well as stabilize the reaction term in the MHD equation.
- The heat transfer rate is enhanced by introducing an obstacle creating turbulence.
- Increase of Reynolds increases the heat transfer but the rate of increase of heat transfer decreases.
- Increase of Hartmann number decreases the heat transfer due to the laminarization of the flow
- At lower turbulence, the rate of heat transfer stabilizes with time whereas in high turbulence i.e for Re 10000 it is still unstable.

The projection based VMS has been utilized for the first time in the MHD problem. But, there is still a lot of scope for the implementation and improvement, such as:

- The Variational multiscale is able to stabilize the flow for time steps more than the simple galkerkin FEM. But, it becomes unstable after $t > 50t_h$. Hence, there is a scope of improvement of VMS so that the model can be simulated for more time-steps.
- SM82 model is not applicable for high Reynolds number. The model starts showing the 3D phenomenon for Re $>$ 10,000. 3D modeling in such case becomes essential. VMS can be used for the actual 3D MHD problem for high Reynolds number.
- For low Reynolds number, the rate of heat transfer via convection becomes constant. To keep varying or increasing the heat transfer effect of moving cylinder can be studied.
- Most of the work is for a long square duct. But in reactors the ducts are not always straight, there is a need to study the effect of the curved duct of different cross-section shape.

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